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Building User Trust in AI Chatbots for Customer Service through Human-Like Cues and Perceived Reliability

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Abstract

This qualitative study explores how human-like cues and system competence shape users' trust and perceptions of reliability in AI-driven chatbot customer service. Data were collected from 28 participants through semi-structured interviews conducted in Pakistan and China. Using thematic analysis supported by NVivo 15, the study identifies key patterns in the formation of user trust and interaction experiences. Two main themes emerged: human-like interaction and emotional connection, and perceived reliability and system competence. The first highlights conversational naturalness, empathy, personalisation, and social presence as drivers of affective trust. In contrast, the second emphasises accuracy, transparency, responsiveness, and data security as core elements of cognitive trust. Together, these dimensions illustrate how emotional and functional factors jointly influence user confidence and satisfaction with chatbots. Beyond reaffirming established trust constructs, the study offers context-specific qualitative insights that deepen understanding of how users in a developing market interpret and negotiate trust in AI-mediated service interactions.

Keywords: Chatbot, Trust, Human-like interaction, Customer service, Thematic analysis

1. Introduction

The rapid advancement of artificial intelligence (AI) has transformed the foundations of marketing and organisational operations, reshaping how value is co-created and delivered [1, 2]. Among the most visible applications of AI are chatbots, computer programs designed to conduct natural language conversations based on organisational data and business rules [3]. Powered by natural language processing (NLP) and machine learning, chatbots are increasingly deployed in customer service to provide real-time, cost-effective, and accessible support. [4].

Empirical evidence suggests that chatbots are progressively replacing human-led interactions in various service domains, including product inquiries, complaint resolution, and purchase support [5]. Their integration across multiple stages of the customer journey highlights their growing strategic role in digital service ecosystems [6]. Major global platforms such as eBay, Amazon, Facebook, and Apple have implemented chatbot systems to manage customer orders, deliver recommendations, and enhance service experiences [7]. According to Chen, Gascó-Hernandez [8], nearly one-third of customer communication leaders have adopted or plan to adopt chatbot technology. While these systems are primarily designed for efficiency and functionality [9]. Customers' perceived service quality depends equally on emotional engagement and relational satisfaction [10].

Customer service remains central to organisational success, as it shapes satisfaction, loyalty, and word-of-mouth behaviour [5]. The global chatbot market is projected to grow from US\$2.6 billion in 2019 to US\$9.4 billion by 2024 [11]. In Pakistan, approximately 27% of adults have interacted with chatbots while shopping, and 40% prefer technology-enabled customer support [5]. Chatbots thus offer mutual value, providing consumers with efficient and engaging experiences while enabling firms to gather data on customer preferences and behaviour [12]. China is now leading the world in the development of AI. China is predicted to invest US\$26.69 billion in AI by 2026, making it the second-largest AI market in the world, according to the Worldwide Artificial Intelligence Spending Guide. The integration of AI-driven chatbots across Chinese customer care systems has increased due to this strong national commitment [13].

However, despite their potential, many users remain hesitant to fully trust chatbot-based interactions. Studies show that customers often perceive chatbots as impersonal, lacking empathy, and less reliable than human

agents [14, 15]. A PointSource survey revealed that only 16% of respondents were fully satisfied with their chatbot interactions, while 27% expressed dissatisfaction [16]. Trust, therefore, emerges as a critical prerequisite for chatbot acceptance and sustained usage [14]. Without trust, chatbots are unlikely to achieve their full potential in enhancing customer experiences.

Despite the growing literature on chatbot adoption, several research gaps remain. First, most existing studies focus on quantitative, technology-centric predictors [17, 18], leaving a limited understanding of the nuanced emotional and relational mechanisms through which trust is formed. Second, while human-like cues are often examined in isolation, little is known about how they interact with system competence factors to jointly shape user trust, especially in real service contexts. Third, research on chatbot trust has been dominated by Western and technologically mature markets, creating a lack of context-specific insights from Asian economies such as Pakistan and China, where digital infrastructures, cultural norms, and trust expectations differ significantly. Finally, few studies explore trust through in-depth qualitative inquiry, limiting our understanding of the micro-processes, tensions, and contradictions users experience during chatbot interactions. To address this gap, the present study explores the key facets influencing user trust in chatbot-based customer service. Specifically, it examines:

RQ1: How do human-like cues in chatbot interactions influence online shoppers' trust in customer service?

RQ2: How do the unique characteristics of chatbot interactions shape users' perceptions of reliability and trustworthiness in service contexts?

Through qualitative inquiry and the six steps of thematic analysis suggested by [19], this study identifies the underlying factors that foster or hinder consumer trust in chatbots. By exploring how humanness cues and perceived reliability contribute to user confidence, this research extends current understanding of AI-driven customer service. It provides actionable insights for businesses seeking to enhance trust and engagement in automated service environments.

The rest of the study is organised as follows: Section 2 examines the theoretical underpinnings and pertinent literature. The methodology is described in Section 3. While the results are presented in Section 4, the analysis and interpretation of these findings are presented in Section 5. Key implications are discussed, practice and policy recommendations are made, and future research topics are identified in Section 6.

2. Literature review

2.1 Theoretical framework: affective and cognitive trust in AI chatbots

This study is theoretically grounded in the affective-cognitive trust framework, which conceptualises trust as comprising emotional and rational dimensions [20]. Affective trust refers to users' emotional bonds, feelings of empathy, and perceived social presence during interactions [21]. In contrast, cognitive trust is based on users' evaluations of system competence, reliability, accuracy, transparency, and data security [22]. While established models such as the Technology Acceptance Model (TAM) primarily focus on perceived usefulness and ease of use to explain technology adoption [23], they provide limited insight into how trust is socially and emotionally constructed in AI-mediated service encounters. Similarly, existing human-computer interaction trust frameworks tend to emphasise usability and interface design without fully capturing users' subjective interpretations of trust [21, 24]. By adopting a qualitative approach, this study extends these models by examining how users actively negotiate affective and cognitive trust in AI chatbot interactions, offering context-specific insights from Pakistan and China.

2.2 Customer service and digital transformation

Customer service plays a vital role in shaping customer satisfaction, loyalty, and brand reputation. Dixon, Freeman [25] describe effective customer service as the ability to efficiently and effectively resolve customer concerns or requests. However, beyond efficiency, exceptional service aims to create positive emotional experiences that delight and engage customers [26]. Prior studies highlight several factors influencing customer service experiences, including managing expectations [27], demonstrating courtesy and empathy [28], and adapting to customers' communication styles [29].

Over the past few decades, customer service delivery has undergone a substantial transformation due to the proliferation of self-service technologies [30]. While traditional service encounters relied heavily on human interactions [31]. Digital solutions, including web portals, mobile apps, and automated systems, have redefined the customer journey. Despite these advancements, customers continue to value responsive and empathetic communication, which can be challenging to replicate in automated settings [32].

Within this evolving landscape, chatbots represent a hybrid form of service, positioned between human-assisted and self-service interactions. Because they simulate natural conversation, chatbots offer customers a more accessible, low-barrier means of engagement compared to static web pages [33]. This positions chatbots as a bridge between automation and personalisation in modern customer service systems. However, existing literature often discusses customer service transformation in broad terms without examining how users emotionally negotiate trust in automated interactions. This gap highlights the need to understand how human-like cues and system performance jointly shape user trust, a central focus of this study.

2.3 Chatbots in customer service

Since the early experiments with conversational programs like *ELIZA* in the 1960s, the concept of chatbots has evolved significantly [34]. Følstad and Brandtzaeg [4] define chatbots as “software-based agents that provide information and services through conversational user interfaces.” While early versions were limited to simple interactions, recent advances in AI and NLP have enabled chatbots to handle complex tasks [35]. Chatbots are now widely used across industries such as healthcare [36], education [37], and corporate support [29]. However, customer service remains their most prominent application area due to their scalability, cost-effectiveness, and 24/7 availability [5].

In this context, chatbots can act either as first-line assistants or as independent service channels that escalate issues to human agents when necessary [4]. Nevertheless, customer perceptions of chatbot interactions vary considerably. While some users appreciate efficiency and immediacy, others find chatbots impersonal or limited in empathy [38]. As a result, the challenge for organisations is to design chatbot systems that balance task performance with human-like interaction quality, fostering both effectiveness and emotional engagement. Although the literature describes the benefits and limitations of chatbot-based services, most studies treat functional and emotional responses separately. Few examine how these reactions interact to shape trust, leaving unclear whether human-like cues can compensate for functional shortcomings, or vice versa. This unresolved tension provides a conceptual starting point for the present study.

2.4 Humanness and anthropomorphism

Human-likeness, or anthropomorphism, is one of the most critical factors influencing customer perceptions of chatbots. Designing chatbots with human-like cues such as conversational tone, empathy, or adaptability can enhance users' sense of social presence and trust [39]. Research shows that when chatbots display social behaviours like self-disclosure, humour, or reciprocity, users report higher satisfaction and emotional connection [29]. According to Khan, Fatima [40], human-like features help customers perceive chatbots as more relatable and trustworthy, fostering a stronger emotional bond with the host company. Similarly, Yen and Chiang [41] observed that chatbots capable of adapting their responses to users' emotional states can enhance perceived authenticity and trustworthiness. Empathy and responsiveness, traits typically associated with human service agents, are particularly influential in forming positive user evaluations [7].

However, the literature remains divided on whether increasing anthropomorphism consistently improves trust. Some research warns that excessive humanness can lead to discomfort or mistrust (the "uncanny valley" effect), while other studies find that human-like cues strengthen emotional connection. This inconsistency reveals a theoretical gap: the need to understand how users interpret and balance human-like cues with expectations of system competence, an issue explored qualitatively in this study.

2.5 Perceived reliability and trustworthiness

Trust in technology-mediated services depends largely on users' perceptions of reliability, accuracy, and competence [42]. For chatbot-based customer service, these dimensions translate into the accuracy of information provided, the credibility of communication, and the competence of the chatbot in resolving user issues [43, 44]. Reliability refers to the chatbot's ability to deliver timely and accurate responses [45]. Credibility involves the user's belief that the chatbot understands their needs and provides honest, relevant information [46]. Competence, on the other hand, relates to the chatbot's efficiency and capability in fulfilling customer requests [12]. Studies suggest that when chatbots demonstrate these qualities, such as up-to-date knowledge, prompt responses, and accurate guidance, users are more likely to perceive them as trustworthy [47].

Given that emerging technologies often carry perceptions of uncertainty and complexity, establishing user trust becomes essential for adoption and continued use [14]. In the context of AI-driven customer service,

trust acts as the foundation that determines whether users view chatbot interactions as credible and reliable substitutes for human assistance [48]. Yet, the literature rarely examines how reliability interacts with human-like cues to influence trust formation. Whether these dimensions operate independently, amplify one another, or conflict remains unclear. This gap underscores the importance of studying trust not as a single construct but as an interplay between affective (human-like) and cognitive (competence-based) factors, precisely the contribution of this research.

3. Methodology

3.1 Research design

This study adopts an exploratory qualitative research design to investigate the underlying facets influencing users' trust in chatbot-based customer service. Given the limited understanding of how human-like cues and perceived reliability shape consumer trust, a qualitative approach is appropriate to capture users' subjective experiences and perceptions in depth [49]. This design allows participants to express their thoughts, emotions, and experiences freely, providing rich insights that may not emerge from quantitative surveys [50]. Accordingly, this study employs thematic analysis by using NVivo 15, a flexible yet rigorous method for identifying, analysing, and interpreting patterns across qualitative data [19].

3.2 Research context

The study was conducted within the context of e-commerce and online retail platforms in Pakistan and China, where AI-driven chatbots are increasingly utilised to facilitate customer service interactions. In Pakistan, the adoption of digital customer service technologies has accelerated, with approximately 27% of adults having interacted with chatbots during online shopping [5]. However, trust in automated systems remains uneven, making Pakistan a valuable context for examining how digital literacy, infrastructural variability, and culturally grounded expectations shape user trust in AI technologies.

In contrast, China represents a more advanced and innovation-driven AI ecosystem, supported by large-scale digitalisation and substantial national investment in artificial intelligence. China is projected to invest US\$26.69 billion in AI by 2026, positioning it as the world's second-largest AI market [13]. The widespread integration of AI-powered customer service tools across

major digital platforms reflects a mature service environment in which consumers routinely interact with chatbots. Examining these two contexts enables a comparative understanding of trust formation in an emerging digital economy (Pakistan) and a technologically advanced market (China), highlighting how affective and cognitive trust evaluations may vary across different levels of technological exposure and digital service maturity.

3.3 Sampling strategy and participants

A purposive sampling strategy was employed to recruit participants who had prior experience interacting with AI chatbots for customer service purposes, such as making product inquiries, tracking orders, or resolving complaints [49, 50]. This sampling technique ensured that participants possessed relevant experiential knowledge to provide meaningful data [51-54]. The inclusion criteria required participants to: (a) have engaged with chatbots in an online retail or service context within the past six months; (b) be at least 18 years old; and (c) be willing to share their experiences voluntarily. A total of 18 and 10 participants were selected from Pakistan and China, respectively, as this number was sufficient to reach data saturation, where no new themes emerged from additional interviews. In qualitative research, Braun and Clarke [19] state that 6-15 interviews are typically sufficient to obtain topic depth; Dworkin [55] and Galvin [56] support this range [54]. The study achieved topic saturation following the 16th interview in Pakistan and the 8th in China, suggesting that the overall sample size was adequate to capture the variety of perspectives [57]. Additionally, heterogeneity in the sample was intentionally sought to capture variations in emotional sensitivity, risk perception, and prior digital exposure, which are known to influence trust formation. Participants represented diverse demographic backgrounds in terms of age, gender, and education level, offering a comprehensive perspective on chatbot trust experiences in the digital service landscape (Table 1).

Table 1: Participants' demographic details

Participant ID	Country	Gender	Age range	Occupation	Usage frequency	Interview mode
P01	Pakistan	Male	20 - 25	Student	Occasionally	Online
P02	Pakistan	Female	26 - 30	Marketing executive	Frequently	Face-to-face
P03	Pakistan	Male	41 - 45	Bank officer	Occasionally	Online
P04	Pakistan	Male	20 - 25	Student	Frequently	Face-to-face
P05	Pakistan	Male	26 - 30	Freelancer	Occasionally	Online
P06	Pakistan	Female	31 - 35	Customer service representative	Frequently	Online
P07	Pakistan	Male	20 - 25	Student	Occasionally	Face-to-face

P08	Pakistan	Female	36 - 40	Entrepreneur	Frequently	Online
P09	Pakistan	Male	26 - 30	Software engineer	Frequently	Face-to-face
P10	Pakistan	Male	31 - 35	School teacher	Occasionally	Online
P11	Pakistan	Female	36 - 40	Sales executive	Occasionally	Face-to-face
P12	Pakistan	Female	20 - 25	Student	Frequently	Online
P13	Pakistan	Male	46 - 50	Banker	Occasionally	Online
P14	Pakistan	Male	31 - 35	Research assistant	Frequently	Online
P15	Pakistan	Female	20 - 25	Student	Occasionally	Online
P16	Pakistan	Male	26 - 30	Customer support agent	Frequently	Online
P17	Pakistan	Male	31 - 35	Digital marketer	Frequently	Face-to-face
P18	Pakistan	Female	36 - 40	Entrepreneur	Occasionally	Online
P19	China	Female	20 - 25	Banker	Frequently	Online
P20	China	Male	41 - 45	Student	Occasionally	Face-to-face
P21	China	Male	31 - 35	Customer support agent	Frequently	Face-to-face
P22	China	Female	31 - 35	Software engineer	Occasionally	Online
P23	China	Female	20 - 25	Freelancer	Frequently	Online
P24	China	Male	36 - 40	Marketing executive	Frequently	Online
P25	China	Male	26 - 30	Lecturer	Occasionally	Face-to-face
P26	China	Male	31 - 35	Entrepreneur	Occasionally	Online
P27	China	Female	36 - 40	Student	Frequently	Online
P28	China	Male	20 - 25	Sales executive	Occasionally	Face-to-face

3.4 Data collection

Data were collected through semi-structured interviews, allowing participants to share their experiences openly while ensuring that discussions remained focused on the study's objectives. The interview guide was developed based on prior literature on chatbot trust, human-likeness, and perceived reliability, and was closely aligned with the research questions (Appendix A). To explore RQ1, participants were asked questions such as: *"How do you feel when a chatbot uses human-like language or expressions during an interaction?"* and *"Do features such as friendliness, tone, or responsiveness affect how much you trust a chatbot?"* To address RQ2, questions included: *"What makes a chatbot seem reliable or unreliable to you?"* *"Can you share a situation where a chatbot's performance increased or decreased your trust in the service?"* *"How do factors such as speed, accuracy, or personalisation influence your trust in chatbot-based customer service?"*

All interviews were conducted in Urdu and Chinese, the participants' native language, to encourage natural expression and enhance the richness of responses [58]. With participants' consent, the interviews were audio-recorded and later transcribed and translated into English for analysis. To ensure accuracy, translations were verified through back-translation and

review by bilingual researchers. To ensure linguistic accuracy and preserve meaning, translations underwent a two-step verification: independent back-translation and cross-checking by bilingual experts. Field notes were kept during interviews to capture tone, hesitation, and emotional cues that may not appear in transcripts. These notes informed coding sensitivity and theme interpretation. Each interview lasted approximately 30-40 minutes and was conducted either face-to-face or via online video calls, depending on participant convenience. Confidentiality and anonymity of all participants were strictly maintained throughout the study. Participants were fully informed that their data would be used exclusively for academic and research purposes. Moreover, the study received ethical approval from the Institutional Review Board of the University of Education, Pakistan (Approval No. UE/DM&AS/UEBS/2025/22). This study was carried out in accordance with all relevant institutional guidelines and the principles of the Declaration of Helsinki.

4. Results

4.1 Familiarising with the data

The first stage of thematic analysis involves deep immersion in the data to develop familiarity with participants' accounts and contextual meanings [59]. In this study, all interview recordings were transcribed verbatim and reviewed multiple times to ensure transcription accuracy and contextual completeness. The lead researcher engaged in repeated readings of the transcripts, making initial margin notes to capture early impressions, recurring patterns, and participant-specific nuances. Selected audio segments were revisited to retain sensitivity to tone, emphasis, and emotional cues that could shape interpretation.

To enhance analytical rigour and coding reliability, the familiarisation process was conducted collaboratively. All members of the research team independently reviewed a subset of transcripts and prepared reflexive memos documenting preliminary interpretations, potential codes, and emerging patterns. These memos formed the basis for regular analytic discussions aimed at developing a shared understanding of the data. Divergent interpretations, particularly those related to culturally embedded expressions, emotionally charged statements, or ambiguous meanings, were openly discussed and resolved through consensus rather than majority rule.

This iterative and reflexive engagement with the data helped establish a common analytical lens before formal coding and reduced the risk of individual researcher bias. By combining independent familiarisation, reflexive memoing, and team-based discussion, the study strengthened the credibility, dependability, and trustworthiness of the subsequent coding and thematic development.

4.2 Generating initial codes

In this stage, interview transcripts were carefully reviewed to identify key ideas related to users' trust in chatbot-based customer service. Using NVivo 15, data were coded inductively to capture recurring patterns such as human-likeness, reliability, responsiveness, and empathy [49]. Each transcript was read multiple times to ensure consistency and accuracy in coding. Table 2 presents brief examples of preliminary codes derived from participants' responses. Although frequency counts are presented to indicate the relative salience of sub-themes, the analysis did not rely on numerical occurrence to determine importance. Consistent with qualitative research principles, themes were interpreted based on their depth, contextual meaning, and contribution to understanding trust formation, rather than on frequency alone. Less frequently mentioned themes sometimes offered equally rich insights into users' trust experiences.

Table 2: Main themes, subthemes and frequency

Main themes	Subthemes	Frequency
Human-like interaction and emotional connection (RQ1)	Conversational naturalness	15
	Empathy and emotional sensitivity	13
	Personalisation of responses	14
	Social presence and tone	11
	Visual and linguistic human cues	9
Perceived reliability and system competence (RQ2)	Accuracy of information	16
	Responsiveness and timeliness	12
	Consistency in interaction quality	11
	Transparency and problem-solving ability	13
	Security and data privacy perception	11

4.4 Searching for themes

Following the initial coding process, the generated codes were carefully examined to identify meaningful patterns and relationships across the dataset. According to Braun and Clarke [19], themes represent essential aspects of the data that capture significant insights related to the research focus. Consistent with Dawadi [60], this stage aimed to uncover recurring patterns that explain how participants perceive trust, reliability, and human-likeness in chatbot interactions. Figure 1 illustrates how all the codes developed in Step 2 were grouped and organised into broader thematic categories, which served as the foundation for theme refinement in the subsequent phase of analysis.

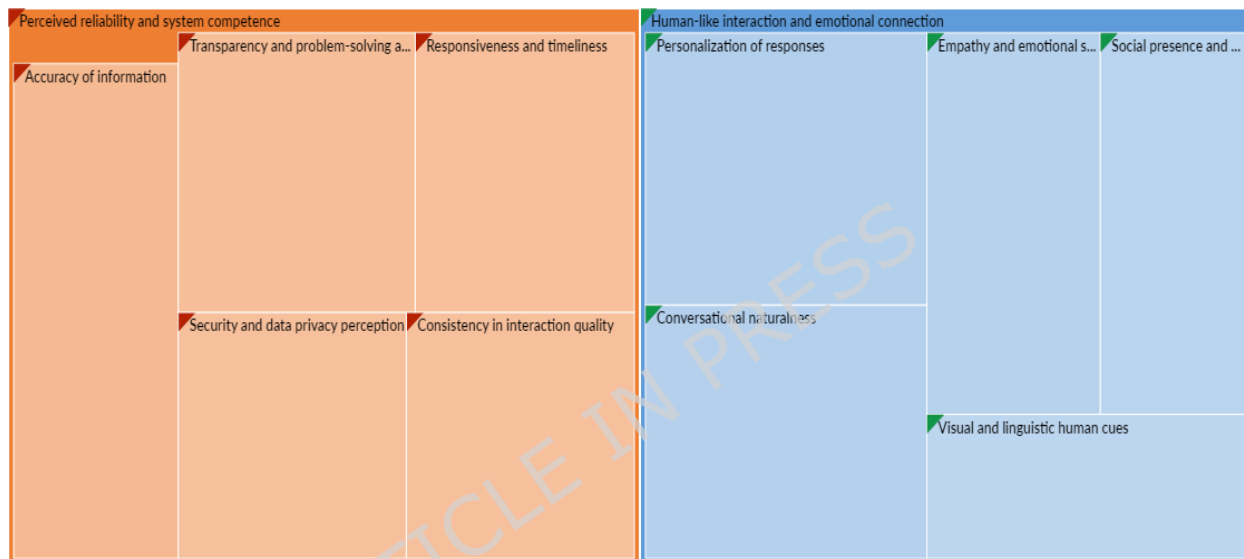


Figure 1: Visualization of themes

4.4 Reviewing themes

In this phase, the preliminary themes were carefully reviewed, refined, and consolidated to ensure they accurately reflected the participants' perspectives. The process involved merging overlapping themes, redefining boundaries, and discarding irrelevant or weak categories [58, 61]. The thematic structure developed in Step 3 was thoroughly evaluated to confirm coherence between the data and the emerging interpretations, ensuring that each theme authentically represented the overall dataset.

4.5 Defining and naming themes

After refining the themes in the previous stage, a clear narrative was developed to capture the essence of each theme and its relevance to the research objectives [49]. At this stage, themes were logically organised to

present a coherent understanding of how participants perceive trust, reliability, and human-likeness in chatbot interactions. Table 3 summarises the finalised and clearly defined themes that emerged from the thematic analysis, representing the key insights derived from participant experiences.

Table 3: Defining themes

Themes	Definition of theme
Conversational naturalness	Refers to the chatbot's ability to engage in smooth, coherent, and human-like conversations that resemble natural human dialogue.
Empathy and emotional sensitivity	Represents the chatbot's capacity to demonstrate understanding, politeness, and emotional awareness in responding to users' needs and sentiments.
Personalization of responses	Indicates the chatbot's adaptation of interactions based on individual user preferences, history, or context, creating a sense of tailored communication.
Social presence and tone	Denotes the perception of human warmth and friendliness conveyed through the chatbot's tone, language style, and conversational flow.
Visual and linguistic human cues	Refers to the use of human-like expressions, such as emojis, greetings, or informal phrasing, that enhance familiarity and relatability.
Accuracy of information	Represents the extent to which the chatbot provides correct, relevant, and reliable information, directly influencing user trust.
Responsiveness and timeliness	Relates to the chatbot's ability to reply quickly and efficiently, minimising waiting time and ensuring smooth communication.
Consistency in interaction quality	Indicates the chatbot's ability to maintain stable performance and uniform communication quality across multiple interactions.
Transparency and problem-solving ability	Refers to the chatbot's openness in communication and its competence in explaining solutions or clarifying limitations when addressing user queries.
Security and data privacy perception	Represents users' confidence that their personal data shared during chatbot interactions is securely managed and protected from misuse.

4.6 Producing the report

The final phase of thematic analysis involved integrating all identified themes into a coherent narrative aligned with the research objectives [19]. Using NVivo 15, the interview data were analysed to highlight how human-like cues and system attributes influence chatbot trust and reliability. A hierarchical coding model (Figure 1) was developed to visually represent the relationships among the main themes and sub-themes [61].

5. Analysis

5.1 Human-like interaction and emotional connection

This study captures participants' perceptions of how chatbots that display human-like qualities foster trust, comfort, and emotional engagement in customer service interactions. The findings suggest that conversational style, empathy, personalisation, tone, and human-like cues collectively shape users' sense of connection with chatbots. Consistent with Nordheim, Følstad [14] and Cai, Gao [62], participants emphasised that the more human-like and emotionally responsive a chatbot appeared, the greater their trust and satisfaction during interactions. These elements not only made the chatbot more relatable but also helped reduce perceived social distance between the user and technology, encouraging continued engagement and loyalty.

5.1.1 Conversational naturalness

Conversational naturalness emerged as a critical factor shaping users' trust and engagement with AI chatbots. Participants indicated that smooth, context-aware dialogue enhanced the perception of authenticity, suggesting that users often interpreted fluency as a proxy for intelligence and attentiveness [63, 64]. However, the data also revealed nuanced tensions. While most participants appreciated coherent, human-like responses, a subset expressed scepticism toward overly polished or excessively "human" language, perceiving it as artificial or potentially manipulative. This indicates that conversational naturalness must strike a balance: sufficient fluency to convey competence and empathy, but restrained enough to avoid uncanny impressions.

Furthermore, the analysis suggests an interrelationship between conversational naturalness and perceived system competence. Users often linked the chatbot's ability to maintain a natural flow with its accuracy and responsiveness; disruptions in conversational coherence, even minor ones, diminished trust in both the emotional and functional dimensions of the interaction [65]. This interplay underscores that naturalness does not operate in isolation; it both signals affective understanding and reinforces cognitive assessments of reliability, highlighting the dual role of conversational quality in shaping user trust. One respondent shared that:

"It felt like chatting with a real person rather than a robot; the responses flowed naturally without sounding mechanical." (P03, P25)

5.1.2 Empathy and emotional sensitivity

Empathy and emotional sensitivity emerged as central drivers of affective trust in AI chatbots. Participants valued chatbots that expressed understanding, politeness, or sympathy, which helped them feel acknowledged and emotionally supported. This aligns with prior research indicating that emotionally responsive chatbots enhance user engagement and foster a sense of care [66, 67]. Emotional sensitivity also mitigated user frustration during problem-solving, effectively humanising the interaction and making minor errors more forgivable. However, the data revealed some tensions. A few participants expressed discomfort with chatbots that appeared “too empathetic” or used scripted emotional responses, perceiving them as insincere or manipulative. This suggests that perceived authenticity is crucial; empathy enhances trust only when users believe the chatbot’s emotional responses are contextually appropriate and not purely artificial.

Interrelationships with other themes were evident. Empathy often amplified the effects of conversational naturalness: users reported that fluent and context-aware dialogue made empathetic responses feel more genuine. Conversely, the impact of empathy was diminished when cognitive trust factors, such as accuracy or responsiveness, were lacking; participants were less likely to value emotional expressions if the chatbot failed to provide reliable or timely information. This highlights the interdependent nature of affective and cognitive trust in AI-mediated interactions. As one interviewee explained:

“When I mentioned I was frustrated, the chatbot apologized politely, which made me feel heard.” (P07, P19)

5.1.3 Personalization of responses

Personalisation emerged as a key mechanism for fostering both emotional and cognitive trust. Participants reported that chatbots that referenced previous interactions, remembered preferences, or used their names created a sense of attentiveness and recognition. This individualised attention enhanced perceptions of human-likeness, engagement, and service quality [68, 69]. Users often interpreted personalised responses as a signal that the system was reliable, competent, and socially aware, strengthening overall trust and willingness to continue interacting. Tensions were evident in user responses. While most participants appreciated personalisation, some expressed concerns about excessive or intrusive personalisation, particularly

when the chatbot recalled sensitive information without clear contextual relevance. Such responses could feel intrusive and reduce trust, suggesting that personalisation must be balanced to avoid the perception of surveillance or manipulation.

Interrelationships with other themes were also observed. Personalisation often amplified the effects of conversational naturalness and empathy; when a chatbot responded naturally and empathetically while referencing prior interactions, users perceived a stronger emotional connection. Conversely, the trust-enhancing effect of personalisation diminished when cognitive factors, such as accuracy or responsiveness, were lacking; participants were less likely to value personalised responses if the chatbot provided incorrect information. This highlights the synergistic interplay between emotional cues and system competence in shaping trust. A participant remarked that:

"It remembered my last order and offered suggestions that made it feel like it actually knew me." (P09, P27)

5.1.4 Social presence and tone

Social presence and tone emerged as pivotal factors influencing users' emotional engagement and trust. Participants emphasised that chatbots employing a friendly, conversational, and polite tone fostered comfort and rapport, encouraging users to share information and complete tasks confidently [67, 70]. Positive phrasing, contextually appropriate humour, and consistent warmth were perceived as signals of attentiveness and relational engagement, reinforcing the chatbot's human-like qualities. The data revealed subtle tensions. While most users preferred approachable language, a minority noted that overly casual or humorous interactions could seem unprofessional or diminish perceived competence. This suggests that tone must be carefully calibrated: sufficient warmth to promote social presence, but constrained to maintain credibility and functional trust.

Interrelationships with other themes were evident. Tone amplified the effects of conversational naturalness and empathy; a natural flow combined with a polite, warm tone strengthened affective trust. Conversely, cognitive trust factors, such as accuracy and responsiveness, moderated the impact of tone; users reported that even a friendly chatbot could not compensate for repeated errors or slow responses. These findings highlight the interdependent nature of emotional and functional trust cues, emphasising

that social presence contributes to trust most effectively when aligned with system competence. One participant expressed that:

“The friendly tone made me comfortable, almost like chatting with a helpful friend.” (P12, P21)

5.1.5 Visual and linguistic human cues

Visual and linguistic human cues, including greetings, emojis, and informal expressions, were identified as important drivers of emotional trust and engagement. Participants reported that such cues made chatbots appear more approachable and relatable, helping to mitigate the impersonal nature of online interactions [71, 72]. These elements contributed to emotional warmth, particularly in retail and service contexts, where users might otherwise feel isolated. The data also revealed nuanced tensions. While most users appreciated these cues as enhancing social presence, some participants perceived excessive use of emojis or informal expressions as unprofessional or distracting. This suggests that the effectiveness of visual and linguistic cues depends on contextual appropriateness and moderation, reinforcing the idea that emotional trust cues must align with users’ expectations of professionalism.

Interrelationships with other themes were evident. Visual and linguistic cues often amplified the effects of conversational naturalness and tone; when greetings or emojis were paired with polite, fluent, and empathetic dialogue, users perceived a stronger human-like presence. Conversely, the impact of these cues diminished if cognitive trust factors, such as accuracy or responsiveness, were compromised. This highlights the interdependent relationship between affective signals and functional competence, emphasising that subtle human-like elements support trust most effectively when the system performs reliably. As highlighted by one respondent:

“Using emojis and greetings made the chatbot feel more lively and human.” (P06, P28)

5.2 Perceived reliability and system competence

This study highlights participants’ trust in chatbots based on their perceived reliability, accuracy, and technical performance during customer service interactions. Users consistently associated trustworthy chatbot behavior with prompt responses, factual accuracy, transparency, and protection of their personal data. In line with Chong, Yu [73] and Naqvi,

Hongyu [5], the findings suggest that when chatbots demonstrate competence and reliability, users perceive them as credible and capable service agents. Participants emphasized that reliable chatbot systems not only enhance confidence in the technology but also reinforce trust in the brand behind it.

5.2.1 Accuracy of information

Accuracy of information emerged as a central determinant of cognitive trust in AI chatbots. Participants emphasised that precise, relevant, and policy-aligned responses were crucial for confidence and continued use. Inaccurate, vague, or inconsistent information quickly undermined trust and prompted users to seek human assistance, highlighting the fragility of cognitive trust when accuracy is compromised [74, 75]. Tensions were observed in user expectations. While most participants valued strict factual correctness, a few indicated that minor errors could be forgiven if the chatbot demonstrated empathy or polite engagement. This underscores the interplay between cognitive and affective trust: emotional cues can mitigate, but not fully offset, the negative effects of low accuracy.

Interrelationships with other themes were also evident. Accuracy reinforced the credibility of human-like cues such as empathy, tone, and personalisation. Participants were more likely to perceive a chatbot's empathetic or personalised responses as genuine when the system consistently delivered accurate information. Conversely, lapses in accuracy weakened the perceived authenticity of emotional cues, suggesting a sequential and interdependent relationship between cognitive competence and affective trust. As one interviewee explained:

"I only trust the chatbot when its information matches what I already know from the website or customer service." (P05, P23)

5.2.2 Responsiveness and timeliness

Responsiveness and timeliness emerged as key determinants of cognitive trust in AI chatbots. Participants highlighted that prompt replies and efficient problem resolution signalled system competence and professionalism, reinforcing confidence in the chatbot as a reliable service agent [76, 77]. Chatbots that minimised waiting times were especially valued compared to traditional customer service channels, enhancing user satisfaction and engagement. However, the data revealed some tensions.

While quick responses were generally appreciated, a few participants noted that excessively rapid or automated replies could feel impersonal or scripted, reducing perceived emotional connection. This suggests a delicate balance between functional efficiency and relational warmth, emphasising the interdependence of affective and cognitive trust cues.

Interrelationships with other themes were also evident. Responsiveness enhanced the credibility of human-like interaction elements, such as empathy and conversational naturalness. Users perceived empathetic or personalised responses as more genuine when delivered promptly. Conversely, delays or repetitive waiting messages weakened both cognitive and affective trust, highlighting that timeliness is foundational to maintaining overall user confidence in AI-mediated interactions. One respondent shared that:

“The quick replies make it convenient; I don’t have to wait like on phone calls.” (P08, P14)

5.2.3 Consistency in interaction quality

Consistency across multiple interactions emerged as a critical factor in shaping cognitive trust and perceived system reliability. Participants reported that coherent and uniform responses signalled a well-designed, stable, and professional system, enhancing confidence in the chatbot and the associated brand [78] [79]. Consistent tone, style, and response patterns reinforced perceptions of technical competence and reliability, contributing to overall user satisfaction. Tensions were observed in instances where participants experienced abrupt changes in tone, contradictory responses, or inconsistent handling of similar queries. Such inconsistencies diminished both cognitive and affective trust, highlighting that even subtle deviations in performance could undermine the perceived authenticity of human-like cues such as empathy or politeness.

Interrelationships with other themes were evident. Consistency reinforced the effectiveness of emotional trust factors: empathetic responses or personalised interactions were considered credible only when the chatbot consistently performed at a high standard. Conversely, inconsistencies in response quality weakened the perceived value of conversational naturalness, social presence, and responsiveness, suggesting a sequential dependency where reliable system performance supports the successful delivery of affective cues. A participant remarked that:

“Every time I used it, the chatbot responded in the same helpful way, which made me confident in using it again.” (P10, P25)

5.2.4 Transparency and problem-solving ability

Transparency and problem-solving ability were identified as key drivers of both cognitive and affective trust. Participants valued chatbots that openly acknowledged their limitations, escalated complex issues to human agents, and clearly explained their functions, rather than pretending to provide complete knowledge [80, 81]. Such transparency was interpreted as honesty and reliability, reinforcing the chatbot’s credibility and reducing uncertainty during interactions. Tensions emerged around user expectations: while transparency was appreciated, some participants noted that excessive disclaimers or frequent escalations could interrupt conversational flow and reduce perceived efficiency. This highlights the need for a balance between openness and seamless interaction to maintain both trust and satisfaction.

Interrelationships with other themes were apparent. Transparent communication amplified the impact of empathy, conversational naturalness, and personalisation; participants were more likely to perceive emotionally sensitive or personalised responses as authentic when the chatbot demonstrated honesty about its limitations. Conversely, lack of transparency diminished the effectiveness of other trust-building cues, even if the chatbot maintained accurate or timely responses, suggesting that transparency serves as a critical moderator between cognitive and affective trust dimensions. As one participant described:

“When the chatbot admitted it couldn’t fix my issue and connected me to a human agent, I felt it was honest and reliable.” (P11, P22)

5.2.5 Security and data privacy perception

Security and data privacy perception emerged as a critical factor shaping users’ cognitive trust in AI chatbots. Participants emphasised the importance of assurances that personal, financial, or sensitive information would be securely handled during interactions. Chatbots that explicitly communicated privacy measures, minimised sensitive data collection, or adhered to transparent data policies were perceived as more trustworthy and reliable [82, 83]. Tensions were noted regarding the trade-off between convenience and privacy. Some participants were willing to provide personal information if it enhanced personalisation or service efficiency, while others

expressed strong reluctance, highlighting diverse user expectations and risk perceptions. This suggests that trust in data handling is not uniform and depends on individual risk tolerance and contextual factors.

Interrelationships with other themes were evident. Perceived security strengthened the effectiveness of affective trust cues, such as empathy and personalisation: users were more likely to respond positively to human-like and emotionally sensitive interactions when they felt their data was protected. Conversely, even highly empathetic or accurate chatbots could lose credibility if privacy concerns were unaddressed, indicating that data security serves as a foundational prerequisite for both cognitive and affective trust. According to a participant:

“If it asks for too much personal info, I hesitate. But when it says my data is secure, I feel more comfortable using it.” (P14, P20)

Figure 2 presents an exploratory conceptual framework developed inductively from the thematic analysis. The framework illustrates how human-like interaction features contribute to the development of affective trust, while system competence factors shape cognitive trust in chatbot-based customer service. These two trust dimensions are conceptualised as sequential and interrelated, whereby affective trust facilitates initial user engagement and openness, and cognitive trust reinforces sustained confidence and continued use. The framework is not intended as a tested causal model but as a theory-building representation that integrates participants' experiences and offers a foundation for future quantitative validation.

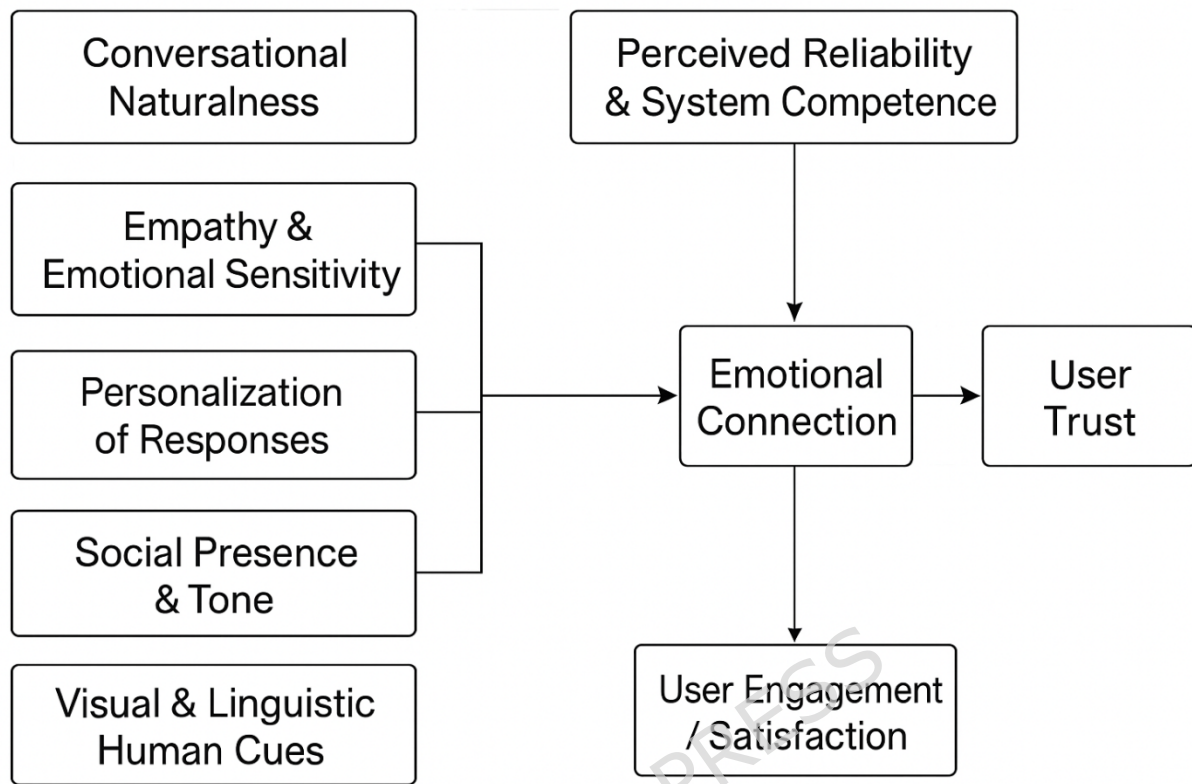


Figure 2: Conceptual framework

6. Discussion

The findings of this study provide valuable insights into how human-like attributes and system competence shape users' trust and engagement with AI-powered chatbots in customer service contexts. Overall, the results reveal that users' trust emerges from a dual foundation, affective trust driven by emotional and social features, and cognitive trust built on perceptions of technical reliability and competence. This aligns with prior research suggesting that both emotional and rational dimensions contribute to user trust in AI-mediated interactions [84, 85].

Participants emphasised that natural conversation, empathy, and personalisation make chatbots appear more "human," leading to stronger emotional engagement and sustained trust. These results reinforce previous studies by Cai, Gao [62] and Tsai, Liu [86], who demonstrated that conversational fluency and social warmth significantly enhance perceived authenticity and comfort. In this study, users described empathetic chatbots as "understanding" and "supportive," reducing frustration and humanising the interaction.

Personalisation emerged as particularly influential in transforming chatbots from mere functional tools into relationship-oriented service agents. This finding supports [68], who highlighted personalisation as a form of digital empathy that promotes user satisfaction and long-term engagement. Likewise, the use of visual and linguistic cues, such as emojis and friendly greetings, fostered a sense of social presence and reduced the emotional gap between users and technology [71].

The second major finding relates to users' perceptions of chatbots as reliable and technically competent service agents. Participants associated accuracy, timeliness, and consistency with a chatbot's professional credibility and reliability. This finding aligns with Naqvi, Hongyu [5] and Chong, Yu [73], who found that reliable performance enhances both user satisfaction and organisational trust. Transparency also played a crucial role in shaping credibility. When chatbots acknowledged their limitations or transferred users to human agents, participants perceived honesty and integrity, reinforcing the argument that algorithmic transparency builds user trust in AI systems. Similarly, data privacy and security assurance emerged as indispensable components of cognitive trust, consistent with Kushwaha, Pharswan [82], who observed that privacy guarantees directly influence user willingness to share personal data.

6.1 Theoretical implications

This study advances theoretical understanding of human-AI trust by extending the affective-cognitive trust framework within the context of AI-driven customer service in emerging digital service environments. While prior trust theories typically conceptualise affective and cognitive trust as coexisting dimensions, this study provides a contextualised explanation of how these dimensions interact and are prioritised by users in developing collectivist markets. Notably, the findings indicate that affective trust often precedes and enables cognitive trust, a sequence that contrasts with dominant Western-centric perspectives in which cognitive evaluations, such as accuracy, usefulness, and system performance, tend to dominate early trust formation.

The theme of *human-like interaction and emotional connection* highlights a culturally grounded reliance on empathy, warmth, personalisation, and conversational naturalness, suggesting that users in low AI-maturity contexts depend more heavily on relational cues to mitigate uncertainty and perceived risk. This finding extends human-computer

interaction trust research by demonstrating that human-like attributes function not merely as interaction enhancers, but as foundational trust signals shaped by cultural expectations for socially oriented communication.

In contrast, the theme of *perceived reliability and system competence* illustrates how transparency, accuracy, responsiveness, and data security act as critical reinforcers of trust once emotional comfort has been established. This dynamic interplay reveals a context-specific trust configuration in which affective trust facilitates initial acceptance, while cognitive trust sustains continued reliance on AI chatbots. Such a sequential and interdependent trust mechanism is not adequately captured by universalist adoption models such as TAM or UTAUT, which tend to privilege rational evaluations and overlook the emotional and cultural processes underpinning trust development.

Collectively, these insights refine existing trust theory by proposing a culturally embedded dual-trust mechanism in which affective and cognitive trust operate not as parallel constructs, but as sequential and mutually reinforcing processes. This theoretical advancement lays the groundwork for future research examining how cultural values, technological exposure, and institutional trust influence human-AI trust dynamics across diverse service contexts.

6.2 Practical implications

The findings of this study offer several practical insights for developers, service managers, and policymakers in Pakistan and China, where AI-driven customer service systems are increasingly embedded in digital service environments. The theme of human-like interaction and emotional connection highlights the value of designing chatbots capable of natural language flow, empathy, and personalisation. In Pakistan, where users often rely on Urdu or mixed-language communication styles, developers should incorporate local linguistic nuances and culturally appropriate expressions to encourage comfort and familiarity. In China, where users are accustomed to advanced AI ecosystems, integrating sophisticated NLP models and sentiment analysis systems can further elevate user experience by enabling chatbots to detect emotional cues and adjust tone accordingly. Across both countries, personalisation, such as remembering customer preferences or prior interactions, can significantly enhance emotional bonding and satisfaction.

Social presence and tone also emerged as powerful contributors to trust. In Pakistan, even subtle human-like elements, such as friendly greetings or relatable conversational styles, can improve perceived authenticity in environments where digital literacy levels vary. In China, where visual and interactive interfaces are more prevalent, expressive avatars, context-aware responses, and platform-specific communication styles (e.g., resembling WeChat service norms) can strengthen the sense of engagement. Organisations in both contexts should ensure tone consistency and culturally localised linguistic styles to enhance user comfort and connection.

Perceived reliability and system competence underscore the need for accuracy, transparency, and data security in chatbot design. Pakistani users often evaluate reliability based on the accuracy of basic information and responsiveness, making real-time support and error-free communication essential. Chinese users, who frequently interact with highly automated systems, expect seamless functionality, rapid problem resolution, and stable cross-platform performance. Clearly communicating a chatbot's capabilities and limitations, and offering smooth escalation to human agents when needed, can enhance transparency and reinforce cognitive trust in both markets.

Data privacy and security remain critical concerns. In Pakistan, where awareness of data protection is increasing but formal regulations are still emerging, service providers must actively reassure users through visible privacy statements and transparent data-handling practices. In China, compliance with evolving national data governance frameworks (such as the Personal Information Protection Law) is essential to maintain trust and regulatory alignment. These requirements highlight the need for cooperation among AI developers, service organisations, and regulators in both countries to establish ethical, secure, and user-centred guidelines for AI-mediated customer service.

By combining affective (emotional) and cognitive (functional) design strategies tailored to each market, organisations operating in Pakistan and China can create chatbot systems that not only address users' informational needs but also deliver emotionally satisfying and trustworthy digital experiences. Such dual-focus design approaches can strengthen customer loyalty, enhance brand perception, and support competitive advantage in rapidly evolving AI-driven service environments.

6.3 Limitations and future research directions

Although this study provides important insights into how human-like cues and system competence shape user trust in AI-driven chatbot interactions, several limitations should be acknowledged, each offering opportunities for more focused future research. First, the qualitative sample of 28 participants, while appropriate for thematic depth, limits the generalizability of the proposed relationships. Future research could quantitatively test the refined conceptual model, particularly examining whether perceived reliability and data privacy concerns act as gatekeepers that condition the effectiveness of human-like cues on trust. This would provide a stronger empirical basis for the proposed moderator-mediator structure.

Second, participants varied in digital literacy, which may have influenced how they interpreted chatbot empathy, personalisation, and conversational naturalness. Future studies could experimentally manipulate levels of conversational fluency or empathy to identify thresholds at which these cues meaningfully shift trust perceptions. For example, a controlled experiment could vary the chatbot's tone or response accuracy to test the boundary conditions under which affective trust outweighs cognitive trust. Third, an intriguing qualitative insight was users' positive response to the transparent admission of limitations. This suggests a promising direction for future research to investigate the "Apology Effect" in AI interactions, exploring how different forms of chatbot transparency (e.g., admitting errors, showing uncertainty, escalating to a human agent) influence forgiveness, trust repair, and continued use.

Finally, this study focused exclusively on customer service chatbots; however, trust dynamics may differ significantly in high-stakes domains such as healthcare, finance, or education. Future researchers could examine domain-specific expectations of empathy, reliability, and data privacy, as well as conduct longitudinal studies to track how repeated interactions, adaptive learning, and personalisation shape users' evolving emotional and cognitive trust over time. Integrating psychological, ethical, and technological perspectives will further enrich theoretical development in human-AI trust.

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